

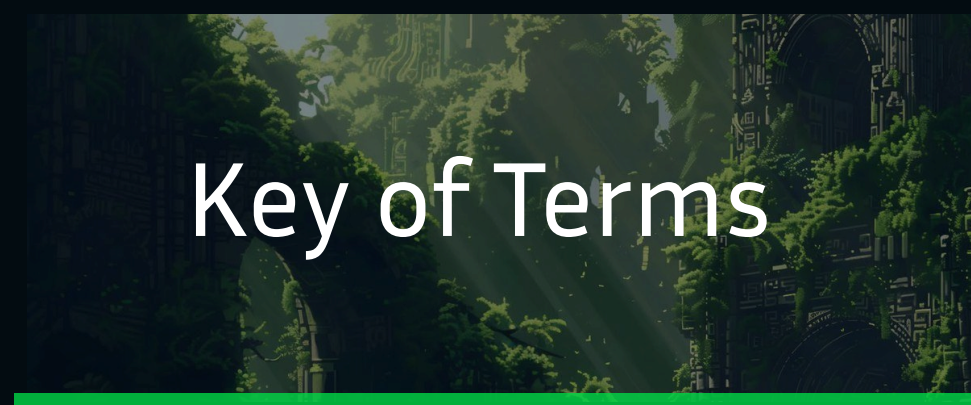
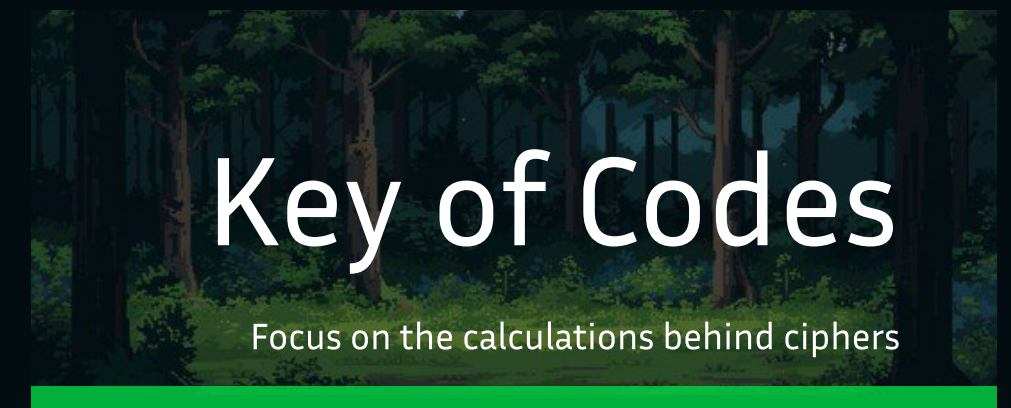


CrypNote teaches cryptography through lessons that unlock keys and knowledge, wrapped in a retro-inspired design.



Start Learning :

Pick a key to explore cryptography step by step. Knowledge awaits behind every lock.





FAQ (Frequently Asked Questions)

Q1: Who is CryptNote For?

A: Anyone interested in learning cryptography, from beginners to advanced learners seeking hands-on challenges.

Q2: Do I need prior programming or math knowledge?

A: Basic familiarity helps, but we start with Fundamentals and provide explanations along the way.

Q3: What are the "Keys" on the dashboard?

A: Each Key is a section of learning: Origins, Secrets, Codes, Terms, References, and Challenges. They guide you from theory to practice.

Q4: Can I solve the challenges without completing all lessons?

A: Yes, but the lessons will give you essential tools and knowledge to tackle challenges efficiently.

Q5: Are the challenges real-world applicable?

A: Absolutely. They are inspired by real CTF problems and cryptographic scenarios, giving you practical experience.

Q6: Is CryptNote Free?

A: The core content is Free, with optional premium Features for advanced challenges and expanded references.

About Us

CrypNote is a learning platform designed to teach cryptography through lessons, challenges, and curated references. Our mission is to make cryptography accessible, interactive, and engaging.



REFERENCE TABLE

Hashing Algorithms Reference

Name	Family	Output Size	Primary Use	Security Status	Notes
MD2	MD	128-bit	Legacy checksum	Broken	Historical only
MD4	MD	128-bit	Legacy	Broken	Severely compromised
MD5	MD	128-bit	Checksums, CTFs	Broken	Fast but collision-vulnerable
SHA-0	SHA	160-bit	Withdrawn prototype	Broken	Never officially used
SHA-1	SHA	160-bit	Legacy Git/certs	Deprecated	Practical collisions (2017)
SHA-224	SHA-2	224-bit	Truncated SHA-256	Secure	Less common variant
SHA-256	SHA-2	256-bit	Bitcoin, TLS, SSL	Secure	Industry standard



REFERENCE TABLE

SHA-384	SHA-2	384-bit	High-security apps	Secure	Truncated SHA-512
SHA-512	SHA-2	512-bit	High-security apps	Secure	Stronger but slower
SHA-512/224	SHA-2	224-bit	Modern alternative	Secure	Better perf than SHA-224
SHA-512/256	SHA-2	256-bit	Modern alternative	Secure	Better perf than SHA-256
SHA3-224	SHA-3 (Keccak)	224-bit	Post-SHA-2 standard	Secure	Different construction
SHA3-256	SHA-3 (Keccak)	256-bit	Post-SHA-2 standard	Secure	Not faster than SHA-2
SHA3-384	SHA-3 (Keccak)	384-bit	Post-SHA-2 standard	Secure	Ethereum uses variant
SHA3-512	SHA-3 (Keccak)	512-bit	Post-SHA-2 standard	Secure	Highest SHA-3 output
BLAKE2b	BLAKE	512-bit	General hashing	Secure	Faster than MD5
BLAKE2s	BLAKE	256-bit	32-bit systems	Secure	Optimized for smaller systems



Cryptography Terms

Cryptography – The science of securing information using mathematical techniques.

Cryptanalysis – The study of breaking cryptographic systems or finding weaknesses in them.

Cipher – An algorithm used to perform encryption or decryption.

Plaintext – Original readable data before encryption.

Ciphertext – Encrypted data that appears unreadable.

Encryption – The process of converting plaintext into ciphertext.

Decryption – The process of converting ciphertext back into plaintext.

Key – A secret value that controls encryption or decryption.

Key Length – The size of the key, usually measured in bits, affecting security strength.

Cryptosystem – A complete system including algorithms, keys, and protocols for secure communication.

Security Objectives :

Confidentiality – Ensuring information is accessible only to authorized parties.

Integrity – Guaranteeing data has not been altered.

Availability – Ensuring systems and data are accessible when needed.

Authentication – Verifying the identity of a user or system.

Authorization – Determining what an authenticated entity is allowed to do.

Non-repudiation – Preventing someone from denying their actions.





Cryptography Terms

Hashing Concepts:

Hash Function – A function that converts data into a fixed-length output.

Digest / Hash Value – The output of a hash function.

Salt – Random data added before hashing to prevent identical outputs.

Pepper – A secret value added to hashing, stored separately from the database.

Collision – When two different inputs produce the same hash.

Preimage Resistance – Difficulty of reversing a hash to find the original input.

Second Preimage – Finding another input that matches a known hash.

Key Derivation Function (KDF) – A method for generating strong keys from weak inputs like passwords.

Symmetric Encryption:

Symmetric Encryption – Encryption using the same key for both encryption and decryption.

Block Cipher – Encrypts fixed-size chunks of data.

Stream Cipher – Encrypts data one bit or byte at a time.

Block Size – The amount of data processed at once in a block cipher.

Mode of Operation – A method that defines how blocks are encrypted.

Padding – Extra data added to fill a block to required length.





Cryptography Terms

Mathematics & Theory

Prime Number – A number divisible only by 1 and itself.

Factorization – Breaking a number into smaller numbers that multiply to it.

Modulo – The remainder after division.

Discrete Logarithm – A hard math problem used in cryptography.

Elliptic Curve – A mathematical structure used in modern public-key systems.

Generator – A value used to create group elements in cryptographic math.

Randomness

Entropy – A measure of unpredictability or randomness.

Pseudorandom – Appears random but is generated algorithmically.

CSPRNG – A cryptographically secure pseudorandom number generator.

Seed – The initial value used to generate random numbers.

Real-World Crypto Usage

TLS – A protocol that secures internet communications.

Handshake – The process of establishing secure parameters.

Session Key – A temporary key used for a single session.

Forward Secrecy – Past communications remain safe even if keys are later exposed.

Man-in-the-Middle (MITM) – An attacker secretly intercepts communication.





Cryptography Terms

CTF / Practical Vocabulary:

Flag – The hidden solution string in a challenge.

Writeup – A documented explanation of how a challenge was solved.

Oracle – A system that reveals information through responses.

Brute Force – Trying every possible key or password.

Dictionary Attack – Using a list of likely passwords.

Exploit – A method that abuses a weakness.

Vulnerability – A flaw that can be attacked.

Data & Encoding:

Encoding – Transforming data into another format for transport or storage.

Hexadecimal – Base-16 representation of data.

Binary – Base-2 representation using 0s and 1s.

Base64 – Encoding binary data into text.

ASCII – A standard mapping of characters to numbers.

Unicode – A universal character encoding system.

Advanced Concepts :

Authenticated Encryption (AEAD) – Encryption that also ensures integrity.

MAC (Message Authentication Code) – Confirms data authenticity using a secret key.

Replay Attack – Re-sending captured data to trick a system.

Side-Channel Attack – Gaining information from physical implementation leaks.

Zero Knowledge Proof – Proving something without revealing the secret.

Homomorphic Encryption – Computing on encrypted data without decrypting it.





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Key of Origins

The History and Foundations of Cryptography

Chapter 1: Ancient Cryptography (3000 BCE - 1400s)

- Why Cryptography Began
- Early Substitution Ideas
- Egyptian Hieroglyph Variations
- Spartan Scytale
- Caesar Shift
- Atbash
- Limits of Ancient Methods
- Lessons From Antiquity

Chapter 2: Medieval & Renaissance Era (1400s - 1700s)

- Rise of Systematic Codebreaking
- Al-Kindi & Frequency Analysis
- Weakness of Monoalphabetic Ciphers
- Polyalphabetic Thinking
- The Vigenère System
- Discovery of Repeating Patterns
- Codebooks & Diplomatic Secrecy
- Lessons From the Era

Chapter 3: Mechanization & Pre-Computer Cryptography (1800s - 1940s)

- Telegraph & Instant Communication
- Commercial Codebooks
- Early Machine Assistance
- The One-Time Pad
- Rotor Machines
- Enigma Design
- Operational Mistakes
- Industrialized Cryptanalysis



Chapter 1: Ancient Cryptography (3000 BCE - 1400s CE)

Why Did Cryptography Begin?

Humans have always needed to keep secrets. Ancient civilizations faced three main challenges:

1. Military Strategy - Commanders needed to send battle plans without enemies intercepting them
2. Trade Secrets - Merchants wanted to protect valuable business information
3. Political Intrigue - Diplomats and rulers needed private communication

The first cryptographers didn't understand mathematics or computers. They just needed practical ways to hide messages.

The Earliest Known Ciphers

1. Egyptian Hieroglyphics (1900 BCE)

What happened: A scribe in Egypt replaced some standard hieroglyphs with unusual symbols in a tomb inscription. This wasn't for security - it was to make the writing seem more mysterious and important.

Key of Secrets

Core Principles and Concepts of Cryptography

Chapter 1 - Foundational Laws of Cryptography

- Kerckhoffs's Principle
- Open Design vs Security Through Obscurity
- Public Review and Peer Analysis
- Historical Failures of Secret Systems
- Trust in Mathematics vs Trust in Secrecy
- What Makes a System Fail

Chapter 2 - Shannon's Framework

Confusion

- Relationship Between Key and Ciphertext
- Non-linearity
- Substitution Mechanisms
- Why Predictability Kills Security

Diffusion

- Spreading Influence
- Avalanche Effect
- Multi-round Transformations

Chapter 3 - Security Goals (CIA+)

- Confidentiality
- Integrity
- Authenticity
- Authorization
- Non-repudiation
- Why Encryption Alone Is Not Enough

Chapter 4 - Symmetric Cryptography

- Shared Secret Model
- Strengths of Symmetric Systems
- Limitations
- Key Distribution Problem
- Where Symmetric Crypto Is Used Today



Chapter 1: The Core Principles

Kerckhoffs's Principle (1883)

The statement: "A cryptosystem should be secure even if everything about the system, except the key, is public knowledge."

What this means: Your security should NOT depend on:

- Keeping the algorithm secret
- Hiding how your system works
- Assuming attackers don't know your methods

Your security SHOULD depend on:

- The secrecy of the key
- The mathematical strength of the algorithm
- Proper implementation

Real-world example:

- DVD encryption (CSS) tried to keep the algorithm secret → broken in 1999
- AES is completely public → still secure 20+ years later

The lesson: Sunlight is the best disinfectant. Public algorithms get scrutinized and improved. Secret algorithms hide flaws.



Key of Codes

Mathematics & Science of Cryptography

Chapter 1 - Arithmetic Engine

The grammar of crypto math.

- Integers & divisibility
- Prime numbers
- Factorization
- Modular arithmetic intuition
- Congruence notation
- Addition & multiplication rules

Chapter 2 - Powers Under Modulo

Where real cryptography begins.

- Exponentiation problem
- Repeated squaring
- Binary decomposition
- Efficiency gains
- Fast power in practice

Chapter 3 - Euclid's Machinery

Why inverses exist.

- GCD
- Coprime meaning
- Euclidean algorithm
- Extended Euclid
- Modular inverse

Chapter 4 - Euler's Hidden Lever

The trapdoor backbone.

- Totient Function
- ϕ For primes
- ϕ For products
- Euler's theorem
- Fermat's special case

Chapter 5 - From Math to Trapdoors



Chapter 1: Number Theory Foundations

Why Number Theory?

Modern cryptography is built on number theory - the study of integers and their properties.

The key insight: Some mathematical operations are easy in one direction but hard in reverse. This asymmetry creates security.

Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31...

Not prime: 4 (2×2), 6 (2×3), 8 (2×4), 9 (3×3), 10 (2×5)...

Why primes matter:

- RSA security depends on difficulty of factoring products of large primes
- Many crypto systems use prime numbers in their construction.

Modular Arithmetic

The clock analogy: 12-hour clock is modulo 12 arithmetic.

$$10 + 5 = 15$$

$$\text{But on a clock: } 10 + 5 \equiv 3 \pmod{12}$$



Key of Challenges

1. Classic Ciphers

- Substitution ciphers: Caesar, ROT13, Atbash
- Transposition ciphers: Rail Fence, Columnar
- Simple monoalphabetic or polyalphabetic ciphers
- Examples: echo "encrypted text" → decrypt manually

2. Modern Cryptography / Crypto Algorithms

- Symmetric encryption: AES, DES, 3DES, ChaCha20
- Asymmetric encryption: RSA, ECC, ElGamal
- Hash functions: MD5, SHA-family, BLAKE2
- HMAC / MAC challenges
- Padding oracle attacks, key recovery, ciphertext manipulation

3. Encoding & Decoding

- Base64, Base32, Base58, Base91
- Hex, binary, URL encoding, uuencode
- Morse code, braille, custom encoding schemes
- Steganographic encoding overlaps sometimes

4. Hash Cracking

- Preimage attacks: find original input from a hash
- Collision attacks: find two inputs producing same hash

Challenge 1 - Caesar Shift

Goal

Recover the original message.

Thinking Path

- Letters only → classical cipher
- Looks like a uniform shift
- Try moving backward in the alphabet



Challenge 2 - Base64

Goal

Decode the text.

Thinking Path

- Mixed case + numbers + = or padding style → encoding, not encryption
- Use base64 decode



Challenge 3 - Single-Byte XOR

Goal

Find the plaintext.

Thinking Path

- Short hex string
- Very common in beginner crypto
- Try XOR with all 256 keys
- Look for readable output



Challenge 4 - Weak Hash Situation

Problem

Two files produce the same MD5 hash.

Goal

What weakness is being demonstrated?



Challenge 5 - RSA Small Exponent